

UNIVERSITY COLLEGE LONDON

EXAMINATION FOR INTERNAL STUDENTS

MODULE CODE : PHAS0042

ASSESSMENT : PHAS0042A6UA
PATTERN

MODULE NAME : Quantum Mechanics

LEVEL: : Undergraduate

DATE : 25 April 2019

TIME : 10:00

TIME ALLOWED : 2 hrs 30 mins

This paper is suitable for candidates who attended classes for this module in the following academic year(s):

Year
2018

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Standard Calculators	Y
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TURN OVER

PHAS0042
Exam paper
Exam 2019

[Tennyson/Monteiro]

Answer ALL SIX questions from Section A and TWO questions from Section B

The numbers in square brackets in the right-hand margin indicate the provisional allocation of maximum marks per question section.

Note the form of the Pauli spin matrices:

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

[Part marks]

Section A

1. Consider a quantum operator $\hat{A}$ and two quantum states Ψ and Φ . Express the following in Dirac notation:

(a) $\psi(x)$. [1]

(b) $\int \Psi^* \hat{A} \Phi d\tau = \int (\hat{A}^\dagger \Psi)^* \Phi d\tau = \left[\int \Phi^* \hat{A}^\dagger \Psi d\tau \right]^*$, where the integrals are over all space. Give corresponding expressions if $\hat{A}$ is Hermitian. [3]

(c) $\int \Psi^* \Phi d\tau$. [1]

(d) Express $\int \Psi^* \Phi d\tau$ using a complete basis $\Psi = \sum_n c_n |\phi_n\rangle$ and $\Phi = \sum_m d_m |\phi_m\rangle$ giving the amplitudes c_n, d_m in terms of other quantities. [3]

2. (a) Show how the Hamiltonian of a quantum system $\hat{H}$ may be represented by a matrix, using a suitable basis of states $|\phi_n\rangle$. [2]
- (b) Outline briefly how the above matrix may be used to obtain the eigenstates $|\chi_j\rangle$ of this quantum system by solving the time-independent Schrödinger equation. [4]

3. If A and B are two observables, represented by operators $\hat{A}$ and $\hat{B}$, then the product of their uncertainties in a given normalised state $|\psi\rangle$ satisfies the inequality

$$\Delta A \Delta B \geq \frac{1}{2} \left| \langle \psi | [\hat{A}, \hat{B}] | \psi \rangle \right|$$

where $(\Delta A)^2 = \langle (\hat{A} - \langle \hat{A} \rangle)^2 \rangle$ and $(\Delta B)^2 = \langle (\hat{B} - \langle \hat{B} \rangle)^2 \rangle$.

Obtain, explaining your reasoning, the uncertainty in measurements for:

- (i) a particle moving in one dimension with position $\hat{x}$ and momentum $\hat{p}$. [3]
- (ii) the angular momenta $\hat{L}_x, \hat{L}_y$ of particle in an eigenstate $|l m_l\rangle$ of $\hat{L}_z$. [3]
4. The unperturbed Hamiltonian $\hat{H}_0$ of a system has non-degenerate eigenvalues $E_n^{(0)}$ and eigenvectors $|\psi_n^{(0)}\rangle$. The system is now perturbed, and the perturbation is described by adding to $\hat{H}_0$ the perturbation Hamiltonian $\lambda \hat{H}'$ where λ is a variable parameter.
- (a) Outline briefly a procedure for correcting the energy eigenvalues E_n and eigenvectors to account for the effect of the perturbation. [5]
- (b) Write the formula for the first-order change $E_n^{(1)}$ of energy eigenvalue E_n . [2]

5. Two spin- $\frac{1}{2}$ particles have spin angular momentum vectors $\hat{S}_1$ and $\hat{S}_2$.

(a) What are the eigenvalues of the corresponding operators $\hat{S}_1^2$ and $\hat{S}_2^2$? [1]

(b) The total spin angular momentum is $\hat{S} = \hat{S}_1 + \hat{S}_2$. What are the two possible values of the square magnitude $\hat{S}^2 = \hat{S}_1^2 + \hat{S}_2^2 + 2\hat{S}_1 \cdot \hat{S}_2$? [2]

(c) For a spin-dependent interaction between the particles, the interaction energy is given by:

$$\hat{H}_{\text{int}} = -\frac{\epsilon}{\hbar^2} \hat{S}_1 \cdot \hat{S}_2 .$$

Hence obtain the energy difference between the singlet and triplet states of the system in terms of ϵ . [4]

6. In quantum theory, particles can be either bosons or fermions.

(a) Explain the meaning of the terms “boson” and “fermion” by referring to spin as well as the symmetry properties of the wave-function describing systems of more than one particle. Give examples of each type of particle. [3]

(b) State the Pauli exclusion principle for fermions, and show how it may be derived by considering the symmetry properties of a two-particle wave-function. [3]

Answer any TWO questions. Only two answers from Section B will be marked.

Section B

7. A quantum state $|\Psi\rangle$ is represented by a complete set of orthonormal basis states, $|\Psi\rangle = \sum_n c_n |\phi_n\rangle$. An alternative representation, using another orthonormal basis is $|\Psi\rangle = \sum_m d_m |\chi_m\rangle$.

(a) Show how a change of representation may be achieved using a matrix S of elements $S_{mn} = \langle \chi_m | \phi_n \rangle$, and a transformation between vectors $d = S c$, carefully explaining your notation. [5]

(b) Explain how S may be used to transform the matrix representation of an arbitrary operator A between two sets of basis states. [5]

(c) A spin- $\frac{1}{2}$ system is characterised by a Hamiltonian:

$$\hat{H} = \epsilon \hbar (\cos \eta \hat{\sigma}_z + \sin \eta \hat{\sigma}_y)$$

Give $\hat{H}$ as a matrix, in the basis $|\alpha\rangle$ and $|\beta\rangle$ of eigenstates of $\hat{\sigma}_z$ and hence obtain its eigenvalues and eigenvectors. [8]

(d) Show that the transformation matrix between the $|\alpha, \beta\rangle$ states and the eigenstates of $\hat{H}$ takes the form:

$$S = \begin{pmatrix} \cos \zeta & -i \sin \zeta \\ \sin \zeta & i \cos \zeta \end{pmatrix}.$$

Suggest a form for S^{-1} and obtain the relation between η and ζ . [6]

(e) For a state of Hamiltonian $\hat{H}$, given above, which is initially $|\Psi(t=0)\rangle = |\alpha\rangle$ obtain a general form for the state at later time t . Hence evaluate $\langle \Psi(t) | \Psi(t') \rangle$, for $t' = t + \pi/\epsilon$. [6]

8. The Hamiltonian of a one-dimensional quantum harmonic oscillator (angular frequency ω , mass of particle m), is expressed as:

$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$. In the operator approach, we consider raising and lowering operators $\hat{a}_{\pm}$ defined in terms of the position and momentum operators $\hat{x}$ and $\hat{p}$ as:

$$\hat{a}_{\pm} = \frac{1}{\sqrt{2}}(\gamma\hat{x} \mp \frac{i}{\hbar\gamma}\hat{p})$$

where $\gamma = \sqrt{\frac{m\omega}{\hbar}}$ and $\langle n|\hat{x}|n\rangle = 0 = \langle n|\hat{p}|n\rangle$ for any eigenstate.

- (a) Show that $[\hat{a}_-, \hat{a}_+] = 1$ and $\hat{H} = \hbar\omega(\hat{N} + \frac{1}{2})$, giving $\hat{N}$ in terms of $\hat{a}_{\pm}$. [8]

- (b) Obtain the effect of the operators $\hat{x}\hat{p}$ and $\hat{p}\hat{x}$ on an eigenstate $|n\rangle$ of $\hat{N}$. You may assume $\hat{a}_-|n\rangle = \sqrt{n}|n-1\rangle$ and $\hat{a}_+|n\rangle = \sqrt{n+1}|n+1\rangle$. [4]

- (c) Hence, for a harmonic oscillator in the state $|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle - \sqrt{\frac{2}{3}}|2\rangle$

- (i) Evaluate $\langle\hat{p}\hat{x}\rangle$ [4]

- (ii) Evaluate $\langle[\hat{x}, \hat{p}]\rangle$. [4]

- (iii) Comment on the above results in relation to the Heisenberg Uncertainty Principle [2]

- (d) Observations on a system with $\gamma = 1$, measure an effective position operator given by:

$$\hat{X}_{\theta} = \frac{1}{\sqrt{2}}[\hat{a}_-e^{i\theta} + \hat{a}_+e^{-i\theta}]$$

where θ is an experimental parameter. Using the results obtained above or otherwise, calculate the product of the uncertainties $\Delta\hat{X}_{\theta}\Delta\hat{X}_{\theta+\pi/2}$ for a harmonic oscillator in the $|n=2\rangle$ eigenstate. [8]

9. (a) Assuming the commutation relations $[\hat{S}_x, \hat{S}_y] = i\hbar\hat{S}_z$ (and all cyclic permutations), show that the raising and lowering operators defined as $\hat{S}_{\pm} = \hat{S}_x \pm i\hat{S}_y$ satisfy $\hat{S}_{\pm}\hat{S}_{\mp} = \hat{S}^2 - \hat{S}_z^2 \pm \hbar\hat{S}_z$. [6]

- (b) Show also that $[\hat{S}_z, \hat{S}_{\pm}] = \pm\hbar\hat{S}_{\pm}$. [4]

- (c) For an arbitrary angular momentum $\hat{S}$, let $|s m\rangle$ denote the simultaneous eigenvector of both $\hat{S}^2$ and $\hat{S}_z$, the corresponding eigenvalue equations being:

$$\hat{S}^2|s m\rangle = \alpha|s m\rangle$$

$$\hat{S}_z|s m\rangle = \beta|s m\rangle.$$

Hence show that the operation of $\hat{S}_+$ on the eigenvalue equation for $\hat{S}_z$ generates a series of eigenstates with eigenvalues equal to $\beta + n\hbar$ where n is an integer. [6]

- (d) An NV colour center in diamond corresponds to an $S = 1$ spin system and is described by the Hamiltonian:

$$\hat{H} = D\hat{S}_z^2 - \mu B\hat{S}_z + C_s(\hat{S}_x^2 - \hat{S}_y^2)$$

where D, C_s are constants and B represents a magnetic field along z . Represent the above Hamiltonian $\hat{H}$ as a matrix. [8]

- (e) Obtain the energy eigenvalues and sketch the energy level structure of the spin system. [3]

Hence obtain an expression for the value of magnetic field B for which the energy levels are equally spaced. [3]

You may find useful the general formula for any angular momentum $\hat{S}$:

$$\hat{S}_{\pm}|s m\rangle = \hbar[s(s+1) - m(m \pm 1)]^{1/2}|s m \pm 1\rangle.$$

10. (a) (i) Explain what is meant by the “variational principle” of quantum mechanics, and write the inequality that expresses this principle in mathematical form, defining the symbols that you use. [6]

(ii) Describe how the variational principle can be used to obtain an approximate value of the ground-state energy of a quantum system. Illustrate your answer by explaining how to apply this method to a real physical system such as the helium atom. [6]

A particle of mass m in one dimension is acted on by the symmetric potential $V(x) = c|x|$, which depends only on the magnitude $|x|$ of the position. The Hamiltonian is:

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + c|x|.$$

The approximate ground-state energy is to be calculated using the normalised trial wave-function $\psi(x) = (\alpha/\pi)^{1/4} e^{-\frac{1}{2}\alpha x^2}$, where $\alpha > 0$ is a parameter that can be varied.

(b) Show that the expectation values of the kinetic energy T and potential energy V with this wave-function are:

$$\langle T \rangle = \hbar^2 \alpha / 4m, \quad \langle V \rangle = c / \sqrt{\alpha \pi}.$$

[6]

(c) Obtain a formula for the value of α at which $\langle H \rangle$ has its minimum value. Hence, obtain an approximate formula for the ground-state energy. [9]

(d) Suggest a strategy for further improving the energy starting from the function which minimised the energy in section 10 (c). [3]

The following integrals may be helpful:

$$\int_0^\infty x e^{-\alpha x^2} dx = \frac{1}{2\alpha}, \quad \int_{-\infty}^\infty (1 - \alpha x^2) e^{-\alpha x^2} dx = \frac{1}{2} \left(\frac{\pi}{\alpha} \right)^{1/2}.$$

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END OF EXAMINATION PAPER